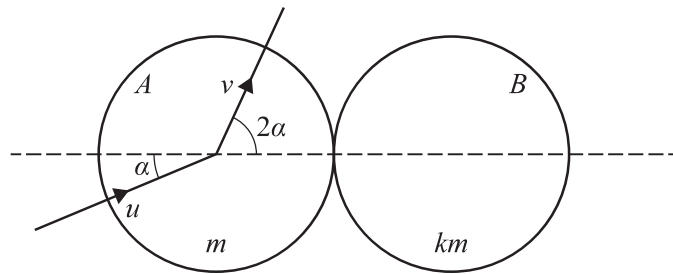




- 5 Two uniform smooth spheres, A and B , of equal radii are on a horizontal surface. They have masses of m and km respectively. Sphere A is moving with speed u at an angle α with the line of centres when it collides with sphere B which is stationary. Immediately after the collision, sphere A moves with speed v at an angle 2α with the line of centres (see diagram).



It is given that $\tan \alpha = \frac{3}{4}$.

- (a) Find v in terms of u . [2]

[illegible]

- (b) Find the coefficient of restitution between the spheres in terms of k . [4]

[illegible]



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(c) Find the range of possible values of k . [3]

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